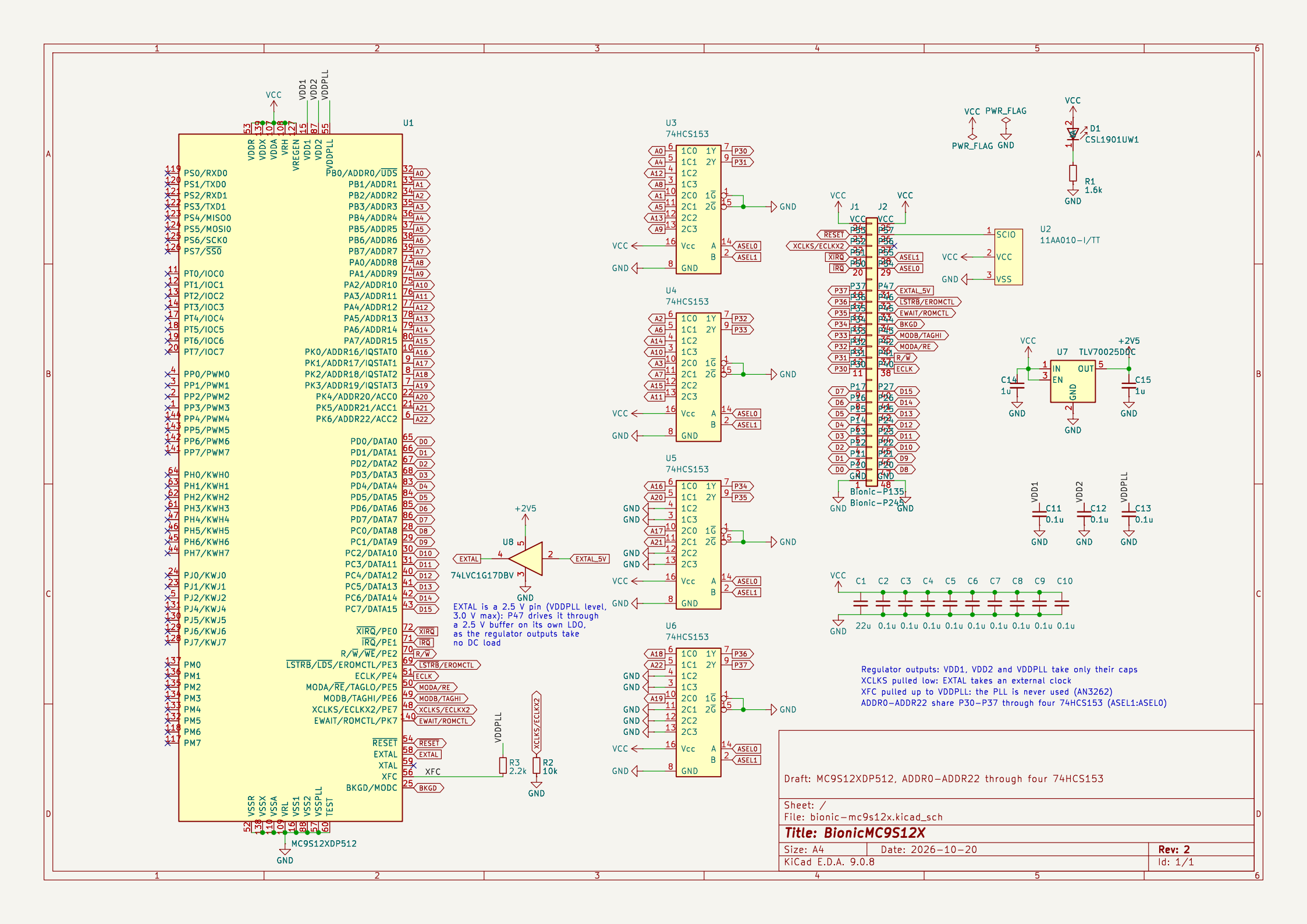

U1
MC9S12XDP512
U3
74HCS153
U4
74HCS153
U5
74HCS153
U6
74HCS153
U8
74LVC1G17DBV
+2V5
EXTAL_5V
EXTAL is a 2.5 V pin (VDDPLL level, 3.0 V max): P47 drives it through a 2.5 V buffer on its own LDO, as the regulator outputs take no DC load
J1
J2
Bionic–P135
Bionic–P245
U2
11AA010–I/TT
U7
TLV70025DDC
D1
CSL1901UW1
R1
1.6k
R2
10k
R3
2.2k
C1 C2 C3 C4 C5 C6 C7 C8 C9 C10
22u 0.1u 0.1u 0.1u 0.1u 0.1u 0.1u 0.1u 0.1u 0.1u
C11 C12 C13
0.1u
C14 C15
1u
PWR_FLAG
Regulator outputs: VDD1, VDD2 and VDDPLL take only their caps
XCLKS pulled low: EXTAL takes an external clock
XFC pulled up to VDDPLL: the PLL is never used (AN3262)
ADDR0–ADDR22 share P30–P37 through four 74HCS153 (ASEL1:ASEL0)
Draft: MC9S12XDP512, ADDR0–ADDR22 through four 74HCS153
Sheet: /
File: bionic–mc9s12x.kicad_sch
Title: BionicMC9S12X
Size: A4
Date: 2026–10–20
Rev: 2
KiCad E.D.A. 9.0.8
Id: 1/1